

This repository contains the datasets of Channel State Information (CSI), Body-coordinate Velocity Profile (BVP) and Doppler Frequency Spectrum (DFS) for human gesture recognition using WiFi.

Cite the Paper

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Readers can also check the homepage of project [Widar3.0](#) for more details about WiFi sensing.

Dataset Description

Data Type	File Name Format	Description
CSI	id-a-b-c-d-Rx.dat	'id': user's id; 'a': gesture type, 'b': torso location, 'c': face orientation, 'd': repetition number, 'Rx': Wi-Fi receiver id.
BVP	id-a-b-c-d-suffix.mat	'id': user's id; 'a': gesture type, 'b': torso location, 'c': face orientation, 'd': repetition number. Each file is a 20*20*T matrix, where the first dimension represents the velocity along x axis ranging between [-2,+2] m/s, the second dimension represents the velocity along y axis ranging between [-2,+2] m/s and the third dimension represents the timestamps with 10Hz sampling rate.
DFS	id-a-b-c-d-suffix.mat	'id': user's id; 'a': gesture type, 'b': torso location, 'c': face orientation, 'd': repetition number. Each file is a 6*121*T matrix, where the first dimension represents the 6 receivers, the second dimension represents the 121 frequency bins ranging between [-60, 60] Hz and the third dimension represents the timestamps with 1000Hz sampling rate.

Note: gesture type refers to [Gesture Description](#) part. Rx location, torso location and face orientation refer to [Device Deployment](#) part. Due to space limit, DFS data can be accessed from our [FTP](#).

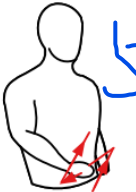
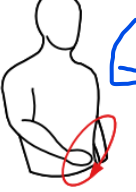
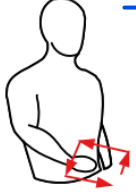
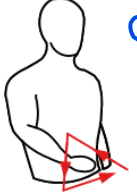
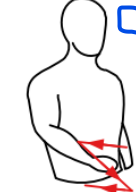
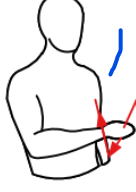
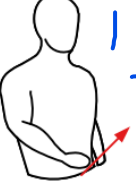
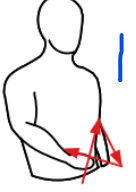
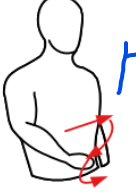
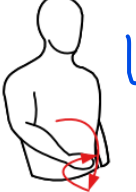
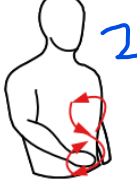
Data Type	Files	Room#	Gestures	Users
CSI	CSI/20181109.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-Zigzag(Vertical); 6: Draw-N(Vertical);	User1,2,3
CSI	CSI/20181112.zip	1	1: Draw-1; 2: Draw-2; 3: Draw-3; 4: Draw-4; 5: Draw-5; 6: Draw-6; 7: Draw-7; 8: Draw-8; 9: Draw-9; 0: Draw-0;	User1,2
CSI	CSI/20181115.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Draw-O(Vertical); 5: Draw-Zigzag(Vertical); 6: Draw-N(Vertical);	User1

Data Type	Files	Room#	Gestures	Users
CSI	CSI/20181116.zip	1	1: Draw-1; 2: Draw-2; 3: Draw-3; 4: Draw-4; 5: Draw-5; 6: Draw-6; 7: Draw-7; 8: Draw-8; 9: Draw-9; 0: Draw-0;	User1
CSI	CSI/20181117.zip	2	1: Push&Pull; 2: Sweep; 3: Clap; 4: Draw-O(Vertical); 5: Draw-Zigzag(Vertical); 6: Draw-N(Vertical);	User4
CSI	CSI/20181118.zip	2	1: Push&Pull; 2: Sweep; 3: Clap; 4: Draw-O(Vertical); 5: Draw-Zigzag(Vertical); 6: Draw-N(Vertical);	User2,3
CSI	CSI/20181121.zip	1	1: Slide; 2: Draw-O(Horizontal); 3: Draw-Zigzag(Horizontal); 4: Draw-N(Horizontal); 5: Draw-Triangle(Horizontal); 6: Draw-Rectangle(Horizontal);	User1,2,3
CSI	CSI/20181127.zip	2	1: Slide; 2: Draw-O(Horizontal); 3: Draw-Zigzag(Horizontal); 4: Draw-N(Horizontal); 5: Draw-Triangle(Horizontal); 6: Draw-Rectangle(Horizontal);	User2,5
CSI	CSI/20181128.zip	2	1: Push&Pull; 2: Sweep; 3: Clap; 4: Draw-O(Horizontal); 5: Draw-Zigzag(Horizontal); 6: Draw-N(Horizontal);	User6
CSI	CSI/20181130_user5_10_11.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal); 7: Draw-N(Horizontal); 8: Draw-Triangle(Horizontal); 9: Draw-Rectangle(Horizontal);	User5,10,11
CSI	CSI/20181130_user12_13_14.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal); 7: Draw-N(Horizontal); 8: Draw-Triangle(Horizontal); 9: Draw-Rectangle(Horizontal);	User12,13,14

Data Type	Files	Room#	Gestures	Users
CSI	CSI/20181130_user15_16_17.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal); 7: Draw-N(Horizontal); 8: Draw-Triangle(Horizontal); 9: Draw-Rectangle(Horizontal);	User15,16,17
CSI	CSI/20181204.zip	2	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal); 7: Draw-N(Horizontal); 8: Draw-Triangle(Horizontal); 9: Draw-Rectangle(Horizontal);	User1
CSI	CSI/20181205.zip	2	[User2]:1: Draw-O(Horizontal); 2: Draw-Zigzag(Horizontal); 3: Draw-N(Horizontal); 4: Draw-Triangle(Horizontal); 5: Draw-Rectangle(Horizontal); [User3]:1: Slide; 2: Draw-O(Horizontal); 3: Draw-Zigzag(Horizontal); 4: Draw-N(Horizontal); 5: Draw-Triangle(Horizontal); 6: Draw-Rectangle(Horizontal);	User2,3
CSI	CSI/20181208.zip	2	[User2]:1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; [User3]:1: Push&Pull;2: Sweep; 3: Clap;	User2,3
CSI	CSI/20181209.zip	2	[User2]:1: Push&Pull; [User6]:1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal);	User2,6
CSI	CSI/20181211.zip	3	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal);	User3,7,8,9
BVP	BVP.zip	All	The gestures correspond to those in CSI data.	All
DFS	DFS.zip	All	The gestures correspond to those in CSI data.	All

Note: The prefix of zip file represents the data collection date. Due to space limit, DFS data can be accessed from our [FTP](#).

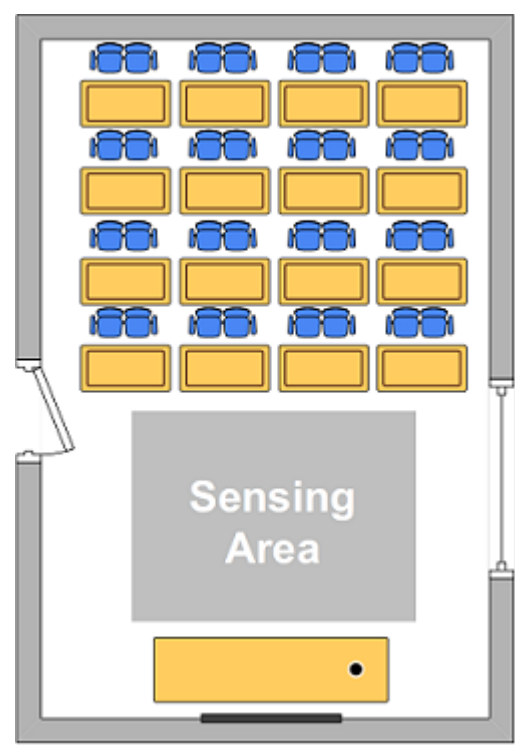
Gesture Description

					
Push & Pull	Sweep	Clap	Slide	Draw-N (H)	Draw-O (H)
					
Draw-Rectangle (H)	Draw-Triangle (H)	Draw-Zigzag (H)	Draw-Zigzag (V)	Draw-N (V)	Draw-O (V)
					
Draw-1	Draw-2	Draw-3	Draw-4	Draw-5	Draw-6
					
Draw-7	Draw-8	Draw-9	Draw-0		

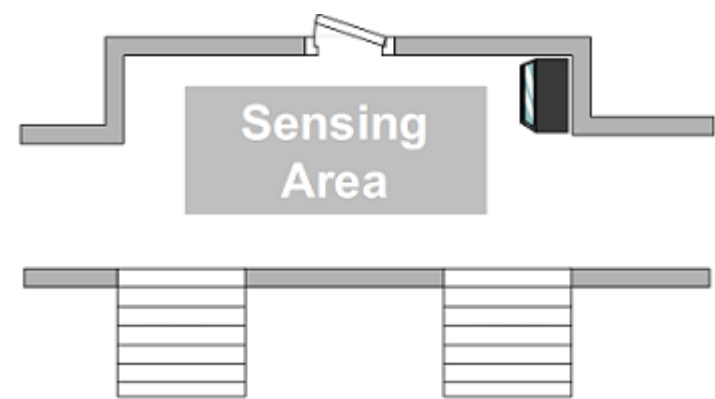
Note: **(H)** represents Horizontal and **(V)** represents Vertical.

Floor Plan

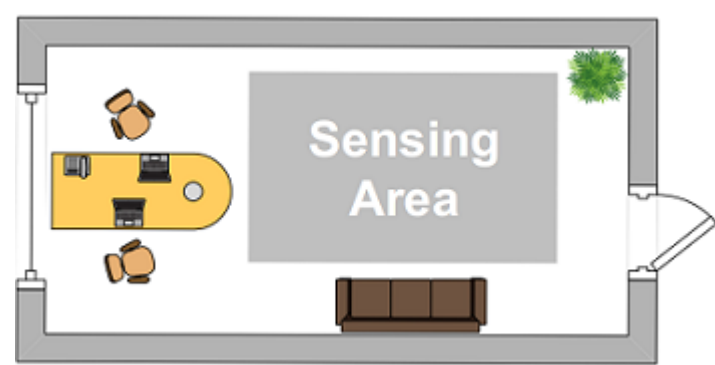
Room#1 - Classroom



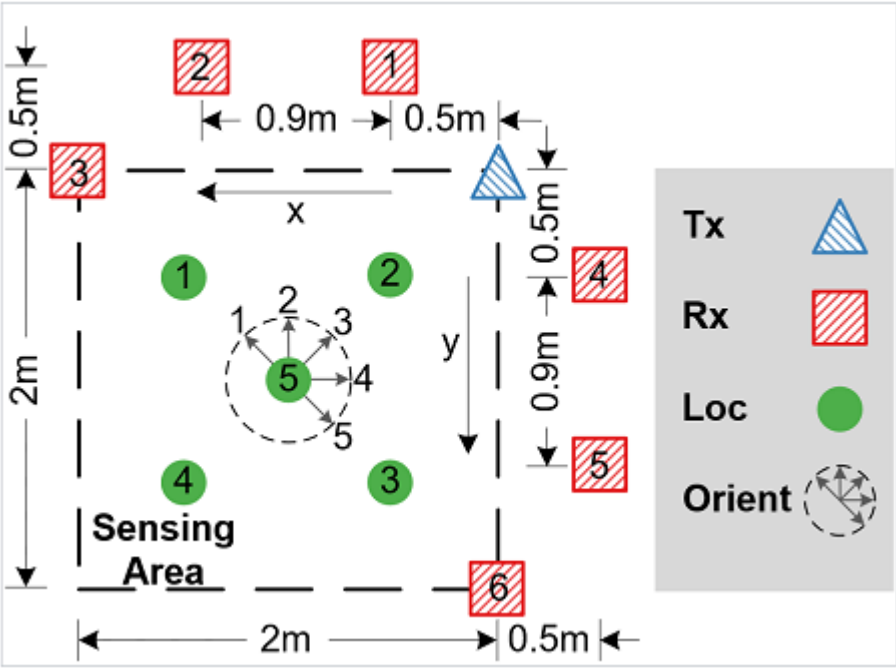
Room#2 - Hall



Room#3 - Office



Device Deployment



Torso Location#	Location (m)	Face Orientation#	Orientation (degree)
1	(1.365, 0.455)	1	-90
2	(0.455, 0.455)	2	-45
3	(0.455, 1.365)	3	0
4	(1.365, 1.365)	4	45
5	(0.91, 0.91)	5	90
6	(2.275, 1.365)		
7	(2.275 2.275)		
8	(1.365 2.275)		

Note: Torso Location #6-8 are not shown in the figure. The reference orientation is denoted as facing the Tx (Orient#3). Tx location is (0, 0).

Code Example

The following shows demo code for BVP extraction and DFS extraction from raw CSI data. Detailed BVP extraction algorithm can be found in our [paper](#).

1. Extract BVP from CSI

```
clear;
clc;
close all;

start_index = [1 1 1 1];
total_mo = 1; % Total motion count
total_pos = 1; % Total position count
```

```

total_ori = 1; % Total orientation count
total_ges = 1; % Total gesture repeation count
start_index_met = 0;
rx_cnt = 6; % Receiver count(no less than 3)
rx_acnt = 3; % Antenna count for each receiver
dpth_pwd = './';
dpth_date = 'Data';
dpth_people = 'userA';
% %%%%%%%%%%%%%%
dpth_ges = [dpth_pwd, dpth_date, '/'];
dpth_vs = [dpth_pwd, 'BVP/'];
for mo_sel = 1:total_mo
    for pos_sel = 1:total_pos
        for ori_sel = 1:total_ori
            for ges_sel = 1:total_ges
                spfx_ges = [dpth_people, '-', num2str(mo_sel), '-',
num2str(pos_sel),...
                        '-', num2str(ori_sel), '-', num2str(ges_sel)];
                if mo_sel == start_index(1) && pos_sel == start_index(2) &&...
                    ori_sel == start_index(3) && ges_sel == start_index(4)
                    start_index_met = 1;
                end
                if start_index_met == 1
                    disp(['Running ', spfx_ges])
                    try
                        DVM_main;
                    catch err
                        disp(['Exception Occured' err.message]);
                        fprintf(exception_fid, '%s\n', spfx_ges);
                        fprintf(exception_fid, '%s\n', err.message);
                        continue;
                    end
                else
                    disp(['Skipping ', spfx_ges])
                end
            end
        end
    end
end
end
end
end

```

The `mo_sel pos_sel ori_sel ges_sel` will traverse all of the CSI file and call `DVM_main` to extract BVP, which is the core algorithm of our paper.

The extracted BVP data will be stored in `/BVP` directory. Make sure you load all of the code component provided in `BVPExtractionCode.zip` in Matlab working space.

2. Extract DFS from CSI

```

clear;
clc;
close all;

```

```

start_index = [1 1 1 1];
total_mo = 1; % Total motion count
total_pos = 1; % Total position count
total_ori = 1; % Total orientation count
total_ges = 1; % Total gesture repetition count
start_index_met = 0;
rx_cnt = 6; % Receiver count(no less than 3)
rx_acnt = 3; % Antenna count for each receiver
dpth_pwd = './';
dpth_date = 'Data';
dpth_people = 'userA';
% %%%%%%%%%%%%%%
dpth_ges = [dpth_pwd, dpth_date, '/'];
dpth_vs = [dpth_pwd, 'BVP/'];

for mo_sel = 1:total_mo
    for pos_sel = 1:total_pos
        for ori_sel = 1:total_ori
            for ges_sel = 1:total_ges
                spfx_ges = [dpth_people, '-', num2str(mo_sel), '-',
num2str(pos_sel),...
                '-', num2str(ori_sel), '-', num2str(ges_sel)];
                if mo_sel == start_index(1) && pos_sel == start_index(2) &&...
                    ori_sel == start_index(3) && ges_sel == start_index(4)
                    start_index_met = 1;
                end
                if start_index_met == 1
                    disp(['Running ', spfx_ges])
                    try
                        [doppler_spectrum, freq_bin] = get_doppler_spectrum(...
                            spfx_ges, rx_cnt, rx_acnt, 'stft');
                    catch err
                        disp(['Exception Occured' err.message]);
                        continue;
                    end
                else
                    disp(['Skipping ', spfx_ges])
                end
            end
        end
    end
end
end
end
end

```

The `mo_sel pos_sel ori_sel ges_sel` will traverse all of the CSI file and call `get_doppler_spectrum` to extract DFS. Both CWT and STFT algorithms are supported. CSI denoising algorithms can be found in our previous works [widar2.0](#) and [widance](#).

The extracted DFS profile is stored in variable `doppler_spectrum` in Matlab working space. You can use it for your custom applications and algorithms. Make sure you load all of the code component provided in *DFSExtractionCode.zip* in Matlab working space.

Bug Notice

List of empty files

- 20181109/user2/user2-6-4-4-2-r1.dat
- 20181109/user3/user3-1-3-1-8-r5.dat
- 20181118/user2/user2-3-5-3-4-r4.dat
- 20181209/user6/user6-3-1-1-5-r5.dat
- 20181211/user8/user8-1-1-1-1-r5.dat
- 20181211/user8/user8-3-3-3-5-r2.dat
- 20181211/user9/user9-1-1-1-1-r1.dat

Please skip these files when loading them into your applications.